

Complexity Science and NSTG-Guided In-Silico Pathology Dynamics for Biologics Pathway Exploration

Computational research thesis manuscript — not a clinical study

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Front matter

Thesis #2

Status. Computational research thesis manuscript. This is not a clinical study, not a medical device dossier, not clinical decision support (CDS), not a dosing advisor, and not a claim of cure.

Repository. <https://github.com/cloudynirvana/complexity-science>

Corresponding objects in this repository. `pathology_cases/` (CaseCard schema and four seed cases); `pipeline/` (NSTG-gated PathwaySketch explorer); `docs/NSTG_PROVENANCE.md`; `DISCLAIMER.md`.

Relation to other work in this repository. Pull request #1 explores a compact multi-scale host–burden ordinary differential equation (ODE) with numeric constraint *hints*. This manuscript describes the complementary **data-first** layer: YAML CaseCards, qualitative Nigeria Standard Treatment Guidelines (NSTG) constraints, and an evidence gate that **never** auto-translates NSTG into ODE coefficients. Numeric scales such as `x_cap_scale` and `infection_risk_weight` are treated here as smuggled non-parameters.

How to read this document. Official NSTG 2022, the Nigeria Essential Medicines List (2020), the National Cancer Control Plan 2018–2022, and the National Policy on Chemotherapy Safety (ChemoSafe, 2021) are **cited, not redistributed**. No guideline chapter, table, or dosing schedule is reproduced. If any line in this repository conflicts with official NSTG 2022, **NSTG wins**. Bibliographic style is Vancouver / NLM (`docs/CITATION_STYLE.md`).

Abstract

Complex pathological systems sit in Warren Weaver’s class of *organised complexity*: many interacting parts, nonlinear couplings, and host context that cannot be reduced to a single rate or a single target [1–3]. In oncology that organisation appears as metabolic competition, immune exclusion, hypoxic and invasive niches, stromal delivery barriers, and latent residual disease [18–26,45–48]. Biologics pathway exploration is therefore a **systemic control problem** — how a host–tumour system might be constrained — not a product-selection problem.

Computational oncology often collapses five distinct epistemic layers into one object. This manuscript treats that collapse as the central failure mode and states a non-negotiable boundary used throughout the accompanying software:

Knowledge ≠ Evidence ≠ Mechanism ≠ Parameter ≠ Prediction

The work reported here is an **architectural** computational thesis, not a clinical study. It specifies, implements, and tests a data-only in-silico pipeline in this repository. A contributor authors a YAML CaseCard — disease framing, systemic axes, research-only observables, candidate mechanisms, falsifiers, qualitative NSTG touchpoints, Vancouver citations, and a required honesty disclaimer. The card is refused if it carries doses, rate constants, ODE knobs, PK/PD tokens, numeric leaves, or NSTG-derived numeric scales. A stub PathwaySketch explorer then ranks bibliographic, falsifiable hypotheses, lists evidence still required, and emits an explicit `refused_non_parameters` record. The evidence gate **never opens**. NSTG 2022 is used only as a structured clinical-knowledge **constraint layer** [49,50]. It is never executed as a care protocol and never auto-translated into model parameters.

Four seed CaseCards were constructed from published knowledge and evidence pointers, not from patient records: triple-negative breast cancer (TNBC) framed as metabolic–immune exclusion [23,28–30]; glioblastoma (GBM) framed as an invasive hypoxic niche [31–34]; pancreatic ductal adenocarcinoma (PDAC) framed as a stromal barrier [37–41]; and dormant / occult residual disease framed explicitly as a research hypothesis space [45–48]. Each card validates against schema version 1.0.0. Across the four cards the explorer emits thirteen ranked hypotheses. Every hypothesis has `parameter_status = refused` and `prediction_status = not_emitted`. Infection- or immunosuppression-themed NSTG touchpoints down-rank immune-axis hypotheses as a qualitative heuristic; the software states that this is not a contraindication and not a parameter. Dedicated tests refuse parameters, `ode_params`, `x_cap_scale`, `infection_risk_weight`, dose-like strings, and bare numeric leaves.

Results are architectural. No ODE was integrated in this pipeline. No public dataset was analysed. No effect size, survival, response, or care pathway is claimed. Named public resources — The Cancer Genome Atlas (TCGA) breast, GBM, and pancreatic cohorts; the NCI Genomic Data Commons; cBioPortal; NCBI GEO; the Ivy Glioblastoma Atlas; and the Human Tumor Atlas Network — are listed only as **future binds** [54–63]. They are not used here.

The contribution is a reusable research object and a closed gate: a CaseCard can be added in under fifteen minutes without editing explorer or ODE code, and NSTG knowledge cannot smuggle itself into a coefficient. That is a complexity-science result about *what must not be computed yet*, not a claim that a biologic pathway has been found.

Keywords

complexity science; organised complexity; in-silico pathology; systemic control; Nigeria Standard Treatment Guidelines (NSTG); knowledge constraint; CaseCard; PathwaySketch; biologics pathway exploration; triple-negative breast cancer; glioblastoma; pancreatic ductal adenocarcinoma; cancer dormancy; parameter smuggling; evidence gate; falsifiability; computational research thesis

1 Introduction

1.1 Complex pathology

A tumour is not a bag of mutated cells. It is a spatially structured, temporally evolving host–tumour system: metabolic sinks, excluded or exhausted lymphocytes, hypoxic niches, desmoplastic stroma, abnormal vasculature, and — after apparent control — occult residual populations [18–26,31,37,45]. Hanahan and Weinberg organised many of these behaviours as hallmarks [18,19]; Hanahan and Coussens, Quail and Joyce, and Junttila and de Sauvage emphasised that recruited stroma and immune context can dominate therapeutic response [20–22]. Joyce and Fearon named T-cell exclusion as a distinct privilege of the tumour microenvironment [23]. Chen and Mellman described the cancer-immunity cycle and the immune set point as systems properties rather than single-receptor facts [24,25]. Hegde and Chen listed exclusion, lack of infiltration, and host context among the practical obstacles to immunotherapy [26].

Those descriptions are knowledge. They are not automatically evidence in a new cohort, not automatically a mechanism in a named patient, not automatically a parameter in a differential equation, and not automatically a prediction of response. Conflating the five is how computational pathology becomes theatre.

1.2 The systemic control problem

Biologics pathway exploration is often narrated as target discovery: find the receptor, write the affinity, rank the product. In a complex host the relevant question is closer to **control under constraint**. Metabolic competition can starve effectors independently of checkpoint occupancy [27,28]. A desmoplastic gland can prevent a small molecule from arriving [37,38]. A hypoxic pseudopalisade can couple survival to migration [32,33]. A dormant disseminated cell can ignore therapies aimed at cycling compartments [45,46]. Infection, anaemia, HIV care continuity, chemotherapy-handling safety, and specialist referral are host and system constraints in the Nigerian guideline and policy environment [49–53]. They shape what may be *explored*. They do not, by themselves, supply a rate constant.

A control problem of this kind is a complexity-science problem. Weaver distinguished organised complexity from problems of simplicity and from disorganised complexity amenable to statistics alone [1]. Anderson’s “more is different” warned that scale introduces new laws rather than larger versions of old ones [2]. Goldenfeld and Kadanoff, and Hartwell and colleagues, restated the lesson for condensed matter and for modular cell biology [3,4]. Kitano’s systems-biology programme, and Barabási, Gulbahce and Loscalzo’s network medicine, made the same claim for disease: the unit of analysis is a coupled system [5–7]. In health services, Plsek and Greenhalgh, Ahn and colleagues, and Lipsitz argued that reduction to a single protocol step misrepresents care [8–10]. None of those arguments licenses an unfitted ODE as a digital twin, and none licenses a guideline as a coefficient.

1.3 What this thesis is

This thesis specifies an in-silico research architecture implemented in complexity-science (repository version 0.2.0, CaseCard schema 1.0.0):

1. a **CaseCard** data schema that holds knowledge, evidence pointers, candidate mechanisms, falsifiers, and qualitative NSTG constraints;
2. a **parameter-smuggling** scanner that refuses numeric and PK/PD/ODE content, including NSTG numeric scales used as constraint hints in a companion ODE pipeline;
3. an **evidence gate** that encodes Knowledge $\neq$ Evidence $\neq$ Mechanism $\neq$ Parameter $\neq$ Prediction and never opens;
4. a stub **PathwaySketch** explorer that ranks hypotheses bibliographically and by falsifiability, applies NSTG theme flags, and records refusals;
5. four **seed cases** — TNBC, GBM, PDAC, dormancy — treated as research objects, not as validated models.

The manuscript reports what the software does on those objects. It does not analyse patient data, does not fit parameters, and does not bind public datasets. It is a thesis about honest computational structure for complex pathology.

1.4 What this thesis is not

This is not a medical device. It is not CDS. It is not dosing advice. It is not a cure. It is not Phase II. It is not regulator-ready. It is not an executable NSTG protocol. It does not speak for the Federal Ministry of Health, Nigeria, or for any manufacturer. Official NSTG text is not redistributed from this repository [49,50].

2 Aims

The aims are computational and architectural.

Aim 1 — Epistemic separation. State and enforce a layered boundary — knowledge, evidence, mechanism, parameter, prediction — so that a biologics pathway hypothesis cannot be written as a coefficient by accident.

Aim 2 — CaseCard as a research object. Define a strict, extra-key-forbidding schema in which a complex pathological case is authored as YAML: disease framing, ≥ 2 systemic axes, research-only observables, ≥ 2 candidate mechanisms each with citations and ≥ 1 falsifier, ≥ 1 qualitative NSTG touchpoint, Vancouver citations with real DOIs for primary and review items, and a required honesty disclaimer.

Aim 3 — NSTG as constraint, not parameter. Use NSTG 2022 and adjacent Nigerian policy documents only as a structured clinical-knowledge constraint layer [49–53]. Cite them. Do not copy them. Do not map them to x_{cap} , infection-risk weights, doses, or rate constants.

Aim 4 — Gated exploration. Implement a PathwaySketch explorer that (i) ranks hypotheses from citations, falsifiers, and qualitative NSTG flags, (ii) lists evidence still required, and (iii) always refuses parameterisation and prediction.

Aim 5 — Seed the hypothesis space. Instantiate four complex cases — TNBC metabolic-immune exclusion, GBM invasive hypoxic niche, PDAC stromal barrier, dormant/occult residual disease — as starting research objects with falsifiers.

Aim 6 — Efficiency without ODE edits. Show that adding a case is a data-only act (typical local cost: under fifteen minutes plus validation) and that the `pathology_cases` and `pipeline` packages do not import `numpy`, `scipy`, or any ODE module.

Non-aims. Fitting ODEs; analysing TCGA, GEO, Ivy GAP, HTAN, or any other public cohort; recommending a product, dose, or schedule; claiming clinical validation; executing NSTG; ingesting patient records.

3 Background

3.1 Complexity science and organised complexity

Weaver’s 1948 essay remains the cleanest statement of the problem this thesis inherits [1]. Problems of *simplicity* have few variables. Problems of *disorganised complexity* have so many weakly coupled parts that averages work. Problems of *organised complexity* have an intermediate number of parts whose interrelations matter. Pathology of the kind encoded in the seed cards is organised: a handful of systemic axes (metabolism, immunity, stroma, hypoxia, invasion, vasculature, dormancy, host) coupled by geometry and history, not by a single stochastic ensemble.

Anderson’s argument that new scales produce new laws [2], and Goldenfeld and Kadanoff’s “simple lessons from complexity” [3], caution against writing a tissue-scale coefficient by renaming a molecule-scale fact. Hartwell, Hopfield, Leibler and Murray recast cell biology in modules rather than lists of genes [4]. Kitano defined systems biology as both a modelling practice and a computational one [5,6]. Barabási, Gulbahce and Loscalzo’s network medicine treats disease as a perturbation of a interactome, not as a one-gene lesion [7]. Holland’s hidden-order programme and Mitchell’s survey of complexity [16,17] add the adaptive piece: the host and the tumour both change the rules while the “experiment” runs.

None of these citations is used here as a licence to simulate a Nigerian patient. They justify a **schema** that keeps host–tumour couplings visible and keeps numbers out until an evidence record — which this stub does not accept — opens a gate.

3.2 Complexity in health systems

Plsek and Greenhalgh described health care itself as a complex adaptive system [8]. Ahn and colleagues asked whether systems biology could discipline medical reductionism [9]. Lipsitz treated health care as a complex system with implications for safety and policy [10]. Those papers are about clinics and organisations. They are relevant here only as a reminder that Nigerian guideline and policy documents are **system-level knowledge**. Referral, handling safety, infection, HIV continuity, anaemia, and palliation are not tumour-cell parameters. A computational pipeline that “implements NSTG” by multiplying a toxicity cap has already made a category error. Pull request #1 in this repository explores numeric constraint hints of that kind as a modelling experiment. This thesis refuses them at the CaseCard boundary so that the data layer cannot be used to launder the error.

3.3 Why model — and when not to

Wolkenhauer asked “why model?” and answered: to make assumptions inspectable [11]. Aldridge, Burke, Lauffenburger and Sorger set out what a physicochemical model of signalling actually requires: a stated network, conservation laws, and parameters with physical meaning, calibrated on targeted experiments [12]. The CaseCard pipeline does not meet those requirements, and it does not pretend to. A YAML hypothesis with two review citations is knowledge-plus-pointer, not a mass-action system.

Ioannidis’s argument that most published findings are false [13] and Begley and Ellis’s demand for higher preclinical standards [14] are used here as **negative** design constraints. A ranked `research_score` that rewards citation count and the presence of a falsifier must not be read as an effect size. Popper’s requirement that a claim name its own destruction [15] is implemented as a schema rule: every candidate mechanism needs at least one falsifier, or the card is rejected.

3.4 Complex pathology as coupled niches

The seed cases are not a convenience sample of “hard cancers.” They are four recurring **control motifs**.

Metabolic–immune exclusion (TNBC). Bianchini and colleagues reviewed TNBC as a heterogeneous disease with immune and residual-disease axes [29]. Li and colleagues reviewed metabolic pathways that exclude or exhaust antitumour immunity [28]. Gatenby and Gillies restated aerobic glycolysis as a population-level strategy, not a single-cell curiosity [27]. Joyce and Fearon separated exclusion from checkpoint occupancy [23]. Schmid and colleagues reported a randomised evaluation of a checkpoint-class antibody plus chemotherapy in advanced TNBC [30]. That trial is **evidence** in its own protocol population. On a CaseCard it is a citation with epistemic: evidence. It is not a coefficient, not a schedule, and not a licence to write 10 mg/kg into YAML — a string the smuggling scanner refuses.

Invasive hypoxic niche (GBM). Hambardzumyan and Bergers defined GBM niches rather than a uniform mass [31]. Brat and colleagues showed that pseudopalisades are hypoxic, protease-expressing, and formed by a migrating population [32]. Giese and colleagues named the migration–proliferation trade-off as a treatment-relevant cost [33]. Semenza reviewed hypoxia-inducible factors as physiology, not as an oncology product class [34]. Bertout, Patel and Simon reviewed oxygen availability in human cancer [35]. Friedl and Wolf reviewed invasion as escape diversity [36]. Vessel count is not delivery; microvascular proliferation can coexist with poor perfusion. That distinction is a hypothesis on the GBM card, not a fitted transport term.

Stromal barrier (PDAC). Kleeff and colleagues reviewed pancreatic cancer as a system-level disease [41]. Feig, Neeße and colleagues reviewed the pancreas-cancer microenvironment and stromal biology [39,40]. Olive and colleagues showed, in a mouse model, that Hedgehog-dependent stroma can limit chemotherapeutic delivery [37]. Provenzano and colleagues showed that enzymatic targeting of hyaluronan-rich stroma can ablate a physical barrier in PDAC models [38]. Kalluri, Sahai and colleagues, and Butcher, Alliston and Weaver extended the stromal and mechanical picture [42–44]. Mouse delivery rescue is **animal evidence**. It is not a human parameter and not an NSTG instruction. The PDAC card says so in its disease notes.

Dormancy / occult residual disease. Aguirre-Ghiso reviewed models and clinical evidence for cancer dormancy [45]. Sosa, Bragado and Aguirre-Ghiso reviewed disseminated-cell dormancy as an awakening field [46]. Massagué and Obenauf reviewed metastatic colonisation [47]. Giancotti reviewed dormancy and reactivation [48]. Occult means below a detection threshold in a given assay. It does not mean absent, and it does not mean a hidden parameter. Awakening cues are research objects. They are not a reason to provoke residual disease in a person. The dormancy card is written so that a future contributor cannot turn “awakening” into a protocol without first failing the honesty and smuggling checks.

3.5 NSTG 2022 as knowledge constraint

The Federal Ministry of Health, Nigeria, published the third edition of the *Nigeria Standard Treatment Guidelines* in 2022 [49]. A Federal Ministry of Information and National Orientation release records an official launch on 24 November 2022 at the Ministry headquarters in Abuja and notes that government communications sometimes use the abbreviation NTSG [50]. This manuscript cites that bibliographic fact. It does **not** quote guideline chapters, tables, or dosing schedules. Official text must be obtained from FMOH or an authorised distributor [49].

Adjacent public policy documents cited on the seed cards, also not redistributed as body text, are:

- Federal Ministry of Health, Nigeria. *Nigeria Essential Medicines List*. 7th edition. 2020 [51].
- Federal Ministry of Health, Nigeria. *Nigeria National Cancer Control Plan 2018–2022*. 2018 [52].
- Federal Ministry of Health, Nigeria. *National Policy on Chemotherapy Safety (ChemoSafe)*. June 2021 [53].

In this architecture NSTG and those adjacent documents are allowed to be **themes**: infection, immunosuppression, HIV, malaria, tuberculosis, anaemia, sickle cell disease, supportive referral, ChemoSafe handling, palliation. The in-repo qualitative catalog (`pipeline/data/nstg_constraint_catalog.yaml`) stores original flags such as `do_not_map_to_rate`, `not_a_dose_cap`, `art_continuity_unmodelled`, `haemoglobin_unobserved`, `referral_not_triage`. It does not store official prose. It does not store `constraint_hints`. If a catalog line disagrees with NSTG 2022, NSTG wins.

What NSTG is not allowed to be here: a parameter source; an executable care protocol; a digital twin of a Nigerian clinic; a reason to withhold or give any product.

3.6 Complementary objects in this repository

Pull request #1 (`complexity_science`) integrates a four-state ODE (burden, immune competence, toxicity, exposure) and uses placeholder research-index summaries with numeric constraint hints (`x_cap_scale`, `infection_risk_weight`, `max_immune_depletion_scale`). That design is a different thesis-shaped experiment: *what happens if* guideline-shaped numbers are allowed inside a toy dynamical system, while still disclaiming clinical use.

This thesis (pull request #2 and the present manuscript) is the complementary refusal. The data-first layer cannot carry those keys. A contributor who pastes `x_cap_scale: 0.8` into a CaseCard is rejected before Pydantic validation finishes. The two designs can coexist in one organisation only if the boundary between them is explicit. The evidence gate is that boundary.

4 Methods

4.1 Study design

This is a computational methods and architecture study. There is no human-subjects protocol, no animal protocol, and no chart review. Inputs are YAML research cards and published bibliographic identifiers. Outputs are validation results, PathwaySketch JSON objects, and this manuscript. The analysis unit is the **card**, not the patient.

Reproducible commands, as shipped:

```
python -m pip install -e ".[dev]"
python -m pipeline validate --all
python -m pipeline explore --case pathology_cases/cases/tnbc_metabolic_immune_exclusion.yaml
python -m pytest
```

Software identity: Python ≥ 3.10 ; pydantic ≥ 2.6 ; pyyaml ≥ 6.0 ; pytest ≥ 7.4 for tests. Package version 0.2.0. Schema version 1.0.0. Sketch version 1.0.0. Licence: MIT. The pathology_cases and pipeline packages are statically checked not to import numpy, scipy, ode, or dynamics.

4.2 CaseCard schema

A CaseCard is the only object a contributor must author. Machine-readable copies live at pathology_cases/schema.py (Pydantic) and pathology_cases/schema/case_card.schema.json (JSON Schema). Extra keys are forbidden (extra = "forbid"). Required fields and their epistemic layer are:

Field	Layer	Constraint
disease	knowledge	Name, optional abbreviations, research framing (≥ 12 characters)
systemic_axes	knowledge	≥ 2 host–tumour couplings; layer in {metabolism, immunity, stroma, hypoxia, vasculature, invasion, dormancy, host}; role in {driver, constraint, context}
observables	evidence pointers	≥ 2 ; research_only must be true

Field	Layer	Constraint
candidate_mechanisms	mechanism	≥ 2 ; each has a hypothesis statement (≥ 24 characters), a class-level biologic_axis, ≥ 1 citation id, optional NSTG touchpoint ids; status is locked to hypothesis
falsifiers	mechanism	≥ 2 ; every mechanism must be covered by ≥ 1 falsifier (if_observed / then_reject)
nstg_touchpoints	knowledge constraint	≥ 1 ; role locked to constraint; qualitative constraint_statement only
citations	knowledge or evidence	≥ 3 ; Vancouver text; primary/review items require a real DOI (10. prefix, no fake / placeholder / example.com); guideline/policy items may omit DOI
disclaimer	honesty	Must contain “not a medical device”, “not dosing”, “not a cure”, and a CDS/clinical-decision-support negation

Identifiers are snake_case. Cross-links are checked: mechanism citation ids and NSTG ids must exist; each NSTG touchpoint must point at a citation; falsifier mechanism ids must exist; every mechanism must have a falsifier. Duplicate ids are refused.

Class-level biologic axes (pipeline/data/biologic_axes.yaml) are families, not products: metabolic_checkpoint, immune_exclusion, immune_checkpoint, stromal_barrier, hypoxia_adaptation, invasive_niche, vascular_delivery, dormancy_maintenance, dormancy_awakening, host_comorbidity.

4.3 Parameter-smuggling refusal

pathology_cases/smuggling.py walks raw YAML before schema validation. Forbidden keys include, among others: parameters, params, ode, ode_params, dose, doses, dosing, dosage, regimen, schedule, pk, pd, ec50, ic50, ed50, kcat, km, kd, kon, koff, clearance, half_life, cmax, cmin, auc, tmax, vmax, y0, dt, solver, rhs, theta, constraint_hints, x_cap, x_cap_scale, infection_risk_weight, max_immune_depletion, max_immune_depletion_scale. The last cluster is the explicit refusal of PR #1-style numeric NSTG hints.

String leaves are scanned for dose-like quantities (10 mg/kg), PK/PD tokens, and schedule tokens (q3w). Citation fields (vancouver, doi, pmid) are exempt from the value scan so that years, volumes, and DOIs remain legal. Any other numeric leaf (integer or float) is refused. The error class is ParameterSmugglingError, wrapped as CaseCardError by the loader.

This is a documentation and schema control, not a legal device.

4.4 NSTG constraint layer

pipeline/nstg_layer.py attaches qualitative statements and catalog flags to a card. It does not emit numbers. Infection-ish themes (infection, hiv, malaria, tb, febrile_neutropenia, immunosuppression) plus immune-class axes (immune_exclusion, immune_checkpoint, dormancy_awakening) produce a ranking **penalty of 2**. The function docstring and the emitted note state that the penalty is not a contraindication and not a parameter.

The catalog loader refuses a file that contains constraint_hints or parameters, or whose role is not constraint_catalog.

Touchpoint constraint_statement fields on the seed cards are original research-index sentences. They name themes (immunosuppression, referral, ChemoSafe, palliation, anaemia, HIV). They do not quote NSTG.

4.5 Knowledge ≠ Evidence ≠ Mechanism ≠ Parameter ≠ Prediction

pipeline/evidence_gate.py implements a permanently closed gate. refuse_parameterization always returns at least three RefusedNonParameter records:

1. source nstg_touchpoints, attempted kind nstg_scale — NSTG will not emit x_cap, infection-risk weights, doses, or other numeric scales;
2. source candidate_mechanisms, attempted kind parameter — hypotheses are not rate constants, EC50-like knobs, or ODE coefficients;
3. source explorer, attempted kind prediction — a PathwaySketch does not predict response, survival, or a care pathway for a person.

A fourth refusal is appended if any touchpoint role is not constraint.

required_evidence_for lists, for each mechanism: independent measurement of named observables that can separate the mechanism from its neighbours; an orthogonal assay before any rate, dose, or ODE coefficient is proposed; the closed-gate reason string; and a concrete test of each linked falsifier. This list is a **debt register**, not a completed analysis.

4.6 PathwaySketch explorer

pipeline/explorer.py is a stub. For each mechanism it computes an integer research_score:

- +2 per supporting citation (labelled “knowledge pointers”);
- +3 if at least one falsifier is linked;
- -2 if the NSTG infection/immunosuppression heuristic applies;
- +1 for validated CaseCard membership, explicitly “not an efficacy credit”.

Blocked mechanisms (no falsifier or no citations) are sorted below hypotheses. Rank is the sort order, not a clinical priority. Each RankedHypothesis locks parameter_status to refused and prediction_status to not_emitted. Sketch notes state: ranking is bibliographic plus falsifiability plus NSTG flags; no ODE was integrated; research_score is not an effect size; no patient record was read.

The explorer never opens the evidence gate. There is no API in this stub that could open it.

4.7 Seed-case construction

Four YAML files were authored in pathology_cases/cases/. Construction rules:

1. Frame a **systemic** motif, not a person and not a care pathway.

2. Use only published reviews and primaries with Crossref-resolvable DOIs for `kind: primary` and `kind: review`.
3. Cite FMOH documents as `kind: guideline` or `kind: policy` without DOI and without pasting their text [49–53].
4. Write falsifiers as observations that would *reject* the mechanism, not as endpoints that would “prove efficacy.”
5. Mark every observable `research_only: true`.
6. Include the standard honesty disclaimer.

DOIs on the seed cards were checked against the Crossref Works API while preparing this manuscript (20 September 2026). All sixteen primary/review DOIs resolved. PMIDs on the cards are digits-only and match the articles as recorded on the cards. No DOI was invented. Guideline and policy items have no DOI field.

No patient-level information was used. No unpublished cohort was used.

4.8 Honesty controls

`pathology_cases/honesty.py` ships the epistemic-boundary string, a short disclaimer, and a non-claims list. Schema validation calls `disclaimer_gaps`. Repository tests assert that README, DISCLAIMER, seed cards, and this manuscript contain “not a medical device”, “not dosing” / “not a cure” language. The explorer copies the short disclaimer, the boundary, and the non-claims onto every sketch.

4.9 What was not done

The following were **not** performed and must not be inferred from the Results:

- numerical integration of any ODE, including the MHBD-4 system in pull request #1;
- parameter estimation, identifiability analysis, or sensitivity analysis;
- survival, response, or receiver-operating-characteristic statistics;
- download or analysis of TCGA, GDC, GEO, cBioPortal, Ivy GAP, HTAN, ICGC, CPTAC, or any other public dataset;
- use of electronic health records or any Nigerian hospital dataset;
- extraction of official NSTG tables;
- recommendation of a biologic product, combination, dose, or schedule;
- a claim of external, prospective, or regulatory validation.

5 Results

Results are **architectural**. They describe research objects and gate behaviour. They do not describe patients, effect sizes, or predicted benefit.

5.1 Four CaseCards validate as research objects

All four seed files load as CaseCard schema 1.0.0. Field counts:

Case id	Systemic		NSTG			
	axes	Observables	Mechanisms	Falsifiers	touchpoints	Citations
tnbc_metabolic_immune_exclusion	2	2	3	3	3	7
gbm_invasive_niche_hypoxia	2	2	3	3	3	7
pdac_stromal_barrier	4	2	3	3	3	8
dormant_occult_disease	4	2	4	4	4	7

Every observable has `research_only: true`. Every mechanism has `status: hypothesis`. Every NSTG touchpoint has `role: constraint`. Every primary/review citation has a Crossref-resolvable DOI. Guideline and policy citations name FMOH documents without embedding their text [49–53].

TNBC axes. Glycolytic competition (driver), T-cell exclusion (driver), host infection context (constraint). Mechanisms: nutrient starvation of effectors; stromal T-cell privilege; checkpoint pathway as a *research class* (explicitly not a product or coefficient). Falsifiers include metabolically competent intra-nest lymphocytes despite a harsh metabolite niche, intra-nest cytotoxic contact despite intact stroma, and persistent exclusion when the relevant coinhibitory pair is absent.

GBM axes. Hypoxic niche (driver), infiltrative front (driver), angiogenic compartment (context), host supportive context (constraint). Mechanisms: hypoxia-coupled adaptive invasion; migration–proliferation trade-off; vessel count is not delivery. Falsifiers include hypoxic but non-migratory palisades, bulk reduction that also clears infiltrate without a motility change, and microvascular density that tracks tracer delivery even under host anaemia.

PDAC axes. Desmoplastic matrix (driver), hypoperfusion (driver), immune access (context), host infection/anaemia (constraint). Mechanisms: Hedgehog-dependent stroma limits delivery; hyaluronan raises a transport barrier; the same barrier excludes immunity. Falsifiers include stromal depletion without a delivery change, hyaluronan-low tumours that remain as impermeable as hyaluronan-high tumours, and intra-nest T cells despite intact desmoplasia.

Dormancy axes. Cellular quiescence (driver), angiogenic pause (context), immune equilibrium (context), host HIV/infection (constraint). Mechanisms: G0/G1-like pause; angiogenic population pause; immune control without clearance; awakening as a research object that is not a protocol. Falsifiers include uniformly cycling residual cells that die under proliferation-targeted pressure, avascular expansion, immune depletion without outgrowth, and a hypothesized awakening cue that leaves cells occult.

These are starting hypotheses with named destruction conditions. They are not validated models.

5.2 PathwaySketch ranking is bibliographic, not predictive

`explore_all()` emitted four sketches. On every sketch the refusal set contained `nstg_scale`, `parameter`, and `prediction`. Every ranked hypothesis had `parameter_status` = `refused` and `prediction_status` = `not_emitted`. Sketch notes stated that no ODE was integrated, that `research_score` is not an effect size, and that no patient record was read.

TNBC sketch. `nutrient_starvation_excludes_effectors` ranked 1 (`research_score` 8). `checkpoint_axis_as_research_class` and `stromal_tcell_privilege` ranked 2 and 3 (score 6). The two immune-axis items received the NSTG infection/immunosuppression down-rank. Six required-evidence items were listed (union across mechanisms).

GBM sketch. `hypoxia_drives_adaptive_invasion` ranked 1 (score 10: three supporting citations). `invasion_proliferation_tradeoff` and `vessel_count_is_not_delivery` ranked 2 and 3 (score 8). No immune-axis penalty applied. Six required-evidence items.

PDAC sketch. The two stromal-barrier mechanisms tied on score 8 and sorted by id: `hedgehog_stroma_limits_delivery` then `hyaluronan_raises_transport_barrier`. `barrier_also_excludes_immunity` ranked 3 (score 6) after the NSTG immune-axis penalty. Six required-evidence items.

Dormancy sketch. `angiogenic_population_pause` and `cellular_g0_pause` ranked 1 and 2 (score 8). `awakening_is_not_a_protocol` and `immune_control_without_clearance` ranked 3 and 4 (score 6) after the infection/immunosuppression penalty. Seven required-evidence items; eight NSTG constraint strings after catalog flags were appended.

Tied scores are broken by mechanism id. That is a deterministic stub rule, not a claim that angiogenic dormancy is “more true” than cellular quiescence.

The immune-axis penalty did what it was designed to do: it kept the hypotheses in the list and marked them. It did not drop them, did not emit a contraindication, and did not write a weight into an equation.

5.3 Parameter-smuggling refusal

Automated tests (`tests/test_parameter_smuggling.py`) demonstrate that clean cards are not the only thing the gate can do. The following mutations of a minimal valid payload are refused:

Mutation	Refused as
<code>top-level parameters: {k: 0.2}</code>	forbidden key / non-parameter
<code>candidate_mechanisms[0].ode_params</code>	forbidden key <code>ode_params</code>
<code>nstg_touchpoints[0].constraint_hints</code> with <code>x_cap_scale</code> and <code>infection_risk_weight</code>	forbidden key <code>constraint_hints</code>
mechanism statement containing 10 mg/kg and q3w	dose-like / schedule token
<code>disease.notes</code> set to the float 0.42	numeric leaf
statement containing EC50	PK/PD token
YAML file containing <code>x_cap: 1.15</code>	forbidden key <code>x_cap</code>
unknown extra key <code>secret_score</code>	extra-key forbid

A card missing a falsifier for a mechanism is rejected by the schema, not by the smuggler. A fake DOI `10.0000/fake-doi` is rejected. A disclaimer that omits the hard non-claims is rejected. An unknown citation id is rejected.

These are positive results about **refusal**. They are the results this thesis is willing to claim.

5.4 The evidence gate stays closed on clean cards

On unmodified seed cards — cards that already contain no numbers — `refuse_parameterization` still emits the three standing refusals (NSTG scale, mechanism-as-parameter, explorer-as-prediction). The gate is therefore doing work even when the author is honest. That is deliberate. A clean card is not a licence to parameterise. Required-evidence strings still name orthogonal assays and the named falsifiers. The closed-gate sentence is attached to every mechanism:

Evidence gate closed: NSTG and CaseCard content are not auto-translated into model parameters. A mechanism is not a coefficient.

No code path in `pipeline/` assigns a float to a rate, a dose, or an initial condition.

5.5 Packages do not import an ODE stack

An AST walk of `pathology_cases/` and `pipeline/` finds no import of `numpy`, `scipy`, `ode`, or `dynamics`. A contributor can add a fifth complex case by copying a seed YAML and filling schema fields. They are instructed not to open `pipeline/` or any ODE file. Local cost, as documented: schema authoring under fifteen minutes; validation in seconds; sketch emission in seconds. Turning NSTG into a coefficient is listed as n/a — refused.

5.6 What the scores are not

If `research_score` were an efficacy credit, TNBC’s metabolic hypothesis would be “better” than its checkpoint hypothesis because it avoided an infection-theme penalty. That reading is forbidden by the sketch notes, the honesty module, this Results section, and the disclaimer. The penalty is a visibility flag for host-context themes. The citation bonus is a completeness flag for bibliographic pointers. The +1 membership bonus exists so that a validated card is distinguishable from an ad-hoc string, not so that a disease is awarded a point of benefit.

6 Discussion

6.1 An architectural answer to a complexity problem

The organised-complexity diagnosis [1–7] is easy to recite and hard to operationalise. The usual operationalisation in computational oncology is to add more states to an ODE or more layers to a network and then to decorate the figure with a guideline logo. This thesis operationalises the diagnosis as a **refusal structure**. The system is allowed to see couplings (systemic axes), allowed to see competing explanations (mechanisms), allowed to see what would kill those explanations (falsifiers), allowed to see host-context themes (NSTG touchpoints), and forbidden to emit the one object everyone wants: a number that looks like a dose or a rate.

That is a control-theoretic move. In a poorly identified system the first job is to stop the controller from acting on a mis-specified plant. The evidence gate is that stop.

6.2 Why NSTG belongs here at all

A complexity-science project that ignored Nigerian guideline and policy context would treat the host as a generic Western trial body. Infection, anaemia, HIV continuity, sickle-cell comorbidity, chemotherapy-handling safety, and specialist referral are not optional decorations on a TNBC or PDAC card written in this repository [49–53]. They are part of the organised complexity.

The failure mode is to honour that context by inventing `infection_risk_weight = 0.4`. A weight is a parameter. A parameter requires evidence that this stub will not pretend to have [12–14]. Qualitative flags — `hiv_theme`, `art_continuity_unmodelled`, `haemoglobin_unobserved` — keep the theme in the sketch and keep the number out. Down-ranking an immune-axis hypothesis under an immunosuppression theme is a way of saying “this coupling is sensitive; do not skip the host.” It is not a way of saying “do not give immunotherapy in Nigeria.” The software is not licensed to say the latter, and the cards are written so that a contributor cannot smuggle the former into an equation.

6.3 Four motifs, one schema

TNBC, GBM, PDAC, and dormancy look like different specialties. Under the schema they are the same kind of object: a small set of systemic axes, a pair of research-only observables, a handful of mutually criticisable mechanisms, and host constraints. That sameness is the efficiency claim. A fifth case — for example another stromal or exclusion motif — does not require a new explorer. It requires a new YAML file that survives smuggling and honesty checks.

The motifs were chosen because they are **control failures of different kinds**. Metabolic exclusion is a resource-competition failure. GBM invasion is a spatial-escape failure. PDAC stroma is a delivery failure. Dormancy is a detection-and-timing failure. A biologics pathway exploration that only knows how to rank targets will miss all four. A pathway exploration that only knows how to integrate an ODE

will be tempted to encode all four as the same four states. The CaseCard keeps the motifs distinct until someone opens a gate this stub does not open.

6.4 Relation to hallmarks, cycles, and set points

Hallmarks [18,19], accessory cells [20], microenvironmental regulation [21,22], exclusion [23], the immunity cycle and set point [24,25], and immunotherapy’s practical challenges [26] are the knowledge background of the seed cards. The cards do not re-derive those reviews. They point at them and then ask for a falsifier. Schmid et al. [30], Brat et al. [32], Olive et al. [37], and Provenzano et al. [38] are the primary-evidence pointers. They remain pointed-at. They are not re-analysed. They are not meta-analysed. A future worker who wants to parameterise any mechanism is told, in the sketch, to bring an orthogonal assay that is not the paper that suggested the mechanism.

6.5 Honesty as a scientific result

It is unfashionable to treat a disclaimer as a result. In this domain the disclaimer is doing scientific work. Ioannidis and Begley–Ellis described environments in which a figure is treated as a fact [13,14]. A PathwaySketch that omitted `refused_non_parameters` would invite that treatment. A CaseCard that accepted `ec50` in a note field would invite it. A catalog that shipped `x_cap_scale` would invite it. The standing refusals on clean cards exist so that a reader of the JSON cannot say “the gate was silent, therefore I may fit.” Silence is not consent.

7 Limitations

Architectural, not empirical. No wet-lab measurement, no imaging study, and no clinical cohort was performed. Seed mechanisms may be incomplete, outdated, or wrong. The falsifiers are design prompts, not executed tests.

Stub ranking. `research_score` is a small integer heuristic. It is sensitive to how many citations an author attached, not to citation quality beyond “has a real DOI.” A poorly conceived mechanism with two reviews outranks a better one with one primary. That is acceptable only because the score is not an effect size and not a care ranking.

NSTG is not ingested. The constraint layer is a short original catalog plus author-written touchpoint sentences. It is not a structured extract of NSTG 2022. It is not complete. It may omit themes that a clinician using the official book would consider essential. Completeness belongs to FMoH, not to this repository [49,50].

Theme matching is lexical. Infection-ish detection is substring matching on touchpoint theme and id. A mistyped theme silently avoids the penalty. A future catalog with richer identifiers would still be qualitative.

No identifiability path. Even if the gate were opened — it is not — the cards do not contain the measurements Aldridge et al. required for a physicochemical model [12]. Observables are named, not quantified.

English-language, oncology-weighted seeds. The four cases are solid-tumour research framings common in the English literature. They are not a map of Nigerian cancer incidence, and they are not a map of infectious-disease complexity except as host constraints.

Companion ODE is out of scope. Results here cannot be read as a validation or a refutation of the MHBD-4 model in pull request #1. They only show that the data layer will not feed that model automatically.

Scholar and indexing limitations. This manuscript is deposited in a public GitHub repository with a PDF and Highwire meta tags. Indexing by Google Scholar or any other service is not guaranteed and is not a claim of peer review.

Single author, computational setting. This is a sole-author computational manuscript. It has not undergone journal peer review at the time of deposit (20 September 2026).

8 Future work

Future work is stated as **binds**, not as analyses performed here. No result in this manuscript depends on the resources below.

8.1 Public datasets as future binds (not analysed)

If a later study opens an evidence record — a new object, not a CaseCard field — the following **named public** resources are the intended first binds. They are listed so that a future author cannot invent a private cohort in the gap. They are not used in Results.

Intended bind	Why it matches a seed motif	Citation for the resource
TCGA breast cohort (TCGA-BRCA) molecular portraits	TNBC heterogeneity as published knowledge, not as a re-analysis here	[54]
TCGA glioblastoma genomic characterisation	GBM core pathways and later atlas work	[55]
TCGA / ICGC pancreatic ductal adenocarcinoma	PDAC genomic context for stromal hypotheses	[56]
TCGA Pan-Cancer project	Cross-tumour comparison of axes, still future	[57]
NCI Genomic Data Commons	Access layer for TCGA and related programs	[58]
cBioPortal	Open exploration of multidimensional cancer genomics	[59]
NCBI Gene Expression Omnibus	Functional-genomics archive for orthogonal expression studies	[60,61]
Ivy Glioblastoma Atlas Project	Anatomic transcriptional atlas of human GBM niches	[62]
Human Tumor Atlas Network	Spatial and temporal tumour transitions at high resolution	[63]
ICGC/TCGA Pan-Cancer Analysis of Whole Genomes	Whole-genome context if a future evidence record needs it	[64]

Rules for any future bind, to be enforced before this repository accepts a parameter object:

1. The bind is a separate evidence record with its own provenance, licence, and split.
2. A CaseCard may *point* at a dataset accession. It may not contain a fitted number from that dataset.
3. NSTG still does not become a coefficient.
4. Negative assays remain detection limits, not proof of clearance (dormancy card).
5. Animal delivery studies remain animal evidence (PDAC card).

No CPTAC, METABRIC, Single Cell Portal, or hospital EHR analysis is claimed. If those resources are used later, they should be named with the same honesty.

8.2 Evidence records and a still-closed default

A future schema may introduce an EvidenceRecord with measured observables, assay identifiers, and an explicit gate: `still_closed` default. Opening the gate should require a human-signed statement that the record is orthogonal to the papers that suggested the mechanism, that no NSTG sentence was parsed into a float, and that no care recommendation will be emitted. Until that object exists, the stub is the specification.

8.3 Richer qualitative constraints

The catalog can grow as original theme lists — malaria, sickle cell, tuberculosis, febrile neutropenia — without becoming a shadow NSTG. Any growth that copies official tables is out of scope and out of licence.

8.4 Explorer versions that remain stubs until identified

Ranking can be replaced by a documented multi-criteria rule (coverage of axes, independence of falsifiers, recency of primaries) without introducing units of mg or h⁻¹. If an ODE is ever coupled, it should consume an EvidenceRecord, not a CaseCard. The AST import ban can remain on `pathology_cases/` even if a sibling package integrates equations.

8.5 More cases, same schema

Additional organised-complexity motifs — for example myeloid exclusion, tertiary lymphoid structure failure, or treatment-induced dormancy — should be new YAML files. They should not be new Python packages.

9 Conclusions

Complex pathology is a systemic control problem in Weaver’s sense of organised complexity [1]. Biologics pathway exploration that jumps from a review article to a rate constant has skipped the problem.

This thesis contributes a data-first in-silico architecture, implemented and tested in this repository, in which:

- a CaseCard holds knowledge, evidence pointers, mechanisms, falsifiers, and qualitative NSTG constraints;
- NSTG 2022 is cited as a knowledge constraint and is never auto-translated into ODE coefficients [49,50];
- four seed cases (TNBC, GBM, PDAC, dormancy) stand as research objects with real, Crossref-checked citations;
- a PathwaySketch explorer ranks bibliographic hypotheses and always refuses parameters and predictions;
- parameter smuggling, including PR #1 numeric scales, is a failing test rather than a hidden feature.

The work is computational research. It is not a medical device, not CDS, not dosing, and not a cure. Named public datasets are future binds, not silent analyses. Official guideline text remains with its publisher.

The honest next measurement is an orthogonal assay against a named falsifier — not a tighter toxicity cap.

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Vancouver / NLM. Policy: docs/CITATION_STYLE.md. Journal items were completed from PubMed MEDLINE (authors, NLM abbreviation, volume, issue, pages, PMID) on 20 September 2026. Printed DOIs were previously resolved on the Crossref Works API the same day; unverified DOIs are not printed. Guideline and policy items are Internet citations with a cited date. Books have no DOI. No DOI was invented.

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Knowledge $\neq$ Evidence $\neq$ Mechanism $\neq$ Parameter $\neq$ Prediction

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